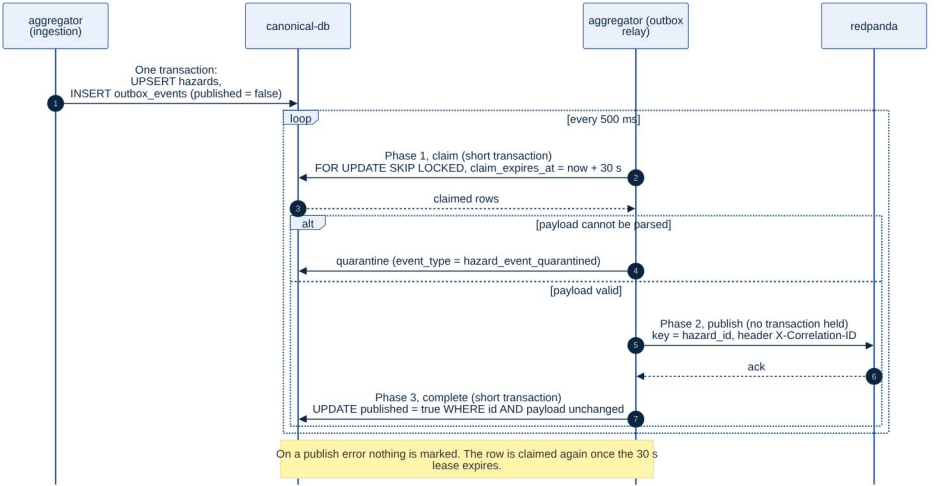

aggregator (ingestion)
canonical-db
aggregator (outbox relay)
redpanda
One transaction:
UPSERT hazards,
INSERT outbox_events (published = false)
1
loop
[every 500 ms]
Phase 1, claim (short transaction)
FOR UPDATE SKIP LOCKED, claim_expires_at = now + 30 s
2
claimed rows
3
alt
[payload cannot be parsed]
quarantine (event_type = hazard_event_quarantined)
4
[payload valid]
Phase 2, publish (no transaction held)
key = hazard_id, header X-Correlation-ID
5
ack
6
Phase 3, complete (short transaction)
UPDATE published = true WHERE id AND payload unchanged
7
On a publish error nothing is marked. The row is claimed again once the 30 s lease expires.